AUTO: before the match

Team 37102 · BioBuzz AUTO RED / BLUE (Pedro) · 30 s, hands off

1 🤝 Talk to partner
Who has S1, who parks in A / B, the GARDEN, our lane (yellow box).

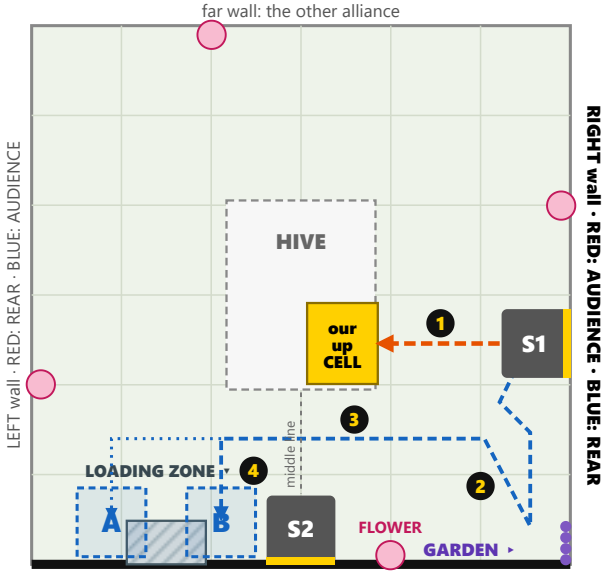
2 📋 Pick op mode
BioBuzz AUTO RED / BLUE = our colour (no other AUTO). TeleOp is preselected.

3 📍 Place + load
Robot at **S1** or **S2** (map). **4 pre-loads**, one in each windmill pocket.

4 ✋ INIT, hands off
Don't touch the robot until START: the Pinpoint calibrates while it is still.

5 🎮 Check + START
Set gamepad 1 buttons, read the screen. ▶ **START** when the match starts.

📍 Where to place it



▲ OUR ALLIANCE WALL · you stand here

Seen from where you stand: the same for RED and BLUE. Yellow bar = intake (front); launcher + camera at the back. **1** shoot · **2** GARDEN · **3** lane beside the HIVE · **4** park in B (dotted: to A)

S1 · CELL wall (default, best): **intake against the wall on your RIGHT** (RED: audience wall, BLUE: rear wall), **centred on our up CELL** (robot middle in line with the CELL middle), square to the wall. It shoots from here.

S2 · ALLIANCE wall: intake against our wall, centred on the middle line (between the FLOWER and the LOADING ZONE), square to the wall. It drives about 5 s to the shot spot, so the GARDEN is short or skipped.

🎮 INIT buttons (gamepad 1)

Press	Changes	Default
D-pad ← or →	Start S1 ⇄ S2	**S1**
□ **Square**	Park slot B ⇄ A	**B**
✕ **Cross**	Shoot pre-loads	**ON**
△ **Triangle**	GARDEN	**ON**
○ **Circle**	Stretch	**OFF**
D-pad ↑ / ↓	Start delay +1 / −1 s	**0 s** (0–10)
R1 / L1	Park by +1 / −1 s	**24 s** (18–27)

✕ △ ○ switch ON ⇄ OFF. One press = one change. **Start delay**: it waits that long after START. **Park by**: the latest time it stops and goes to park. **Stretch**: shoot again after the GARDEN (page 2). The screen calls L1 / R1 "LB / RB". **Every INIT starts from these defaults**: after a new INIT, set them again.

🤝 Agree with your partner (every match)

1. **Start:** only one robot can use S1.
2. **Park:** one robot in A, one in B (the screen says `we park in slot B, they take slot A`). They park **only at their end** of the zone, close to the wall.
3. **GARDEN:** we take our GARDEN POLLEN. They should not.
4. **Lane:** at the end of AUTO they stay out of our lane beside the HIVE (map, 3).

B is the default: the way to A goes past B.

📺 INIT screen: a good one

```
BIOBUZZ AUTO RED - Pedro                    our colour?
Start (D-pad L/R) : CELL wall: side wall in front of our up   as agreed
CELL (shoots from here)
Park slot (Square/X) : B = MIDDLE end of the LOADING          as agreed
ZONE (toward the FLOWER)
>> Tell your partner: we park in slot B, they take slot A
Shoot pre-loads (Cross/A) : ON (1440 t/s)                     ON
GARDEN POLLEN (Triangle/Y) : ON                               ON
Stretch: shoot again (Circle/B) : OFF
Start delay (D-pad U/D) : 0 s
Park by (LB/RB) : 24 s at the latest
Plan : shot spot + 4 pre-loads done ~6 s
Plan : GARDEN done ~17 s (intake on, 2 in back and forth ...)
Plan : PARK B starts by 23 s, takes ~6 s
Route check : OK                                              must be OK
Pedro : ready                                                 must be ready
Pinpoint : READY                                              READY
Battery : 12.9 V                                              12.5 V or more
Camera : our CELL RED_AUDIENCE UP: height 46 in               UP
Place: intake against the AUDIENCE wall, centred on           robot there?
our up CELL, square to the wall
```

RED, all defaults (BLUE: REAR wall). Camera "not in view" is OK too.

Bad line	What happens · what you do
`!! BATTERY 12.1 V …`	Put in a charged battery (12.5 V+).
`!! OUR CELL … DOWN`	Won't shoot (G417). Practice: reset the HIVE.
`!! PEDRO NOT READY …`	FALLBACK: S1 still shoots, then LEAVE. No GARDEN, no PARK. Tell the coach.
`Camera : NOT FOUND …`	Fires unaimed, no stretch. Tell the coach.
`GARDEN … needs shooting`	Shoot is OFF, so no GARDEN (G407). ✕ = Shoot ON.
`no "GARDEN done" Plan line`	No time: it skips the GARDEN.

AUTO: in the match

what the robot does · hands off · if something goes wrong · checklist

What the robot does by itself (30 s)

1 0–1 s
LEAVE
Leaves the wall, spins up the flywheel.
LEAVE +3 (counted at the end)

2 1–7 s
SHOOT
Camera aims at our up CELL, fires the 4 pre-loads.
HIVE TIP +20 if it tips

3 7–14 s
GARDEN
Sweeps slowly along the wall into the GARDEN corner, intake on, until it touches the wall.
POLLEN for TeleOp

4 14–18 s
CORNER
Intake on up to 4 s: backs 2 in off the wall and pushes in again, so the last POLLEN rolls in.

5 ~18 s
PARK
In our LOADING ZONE. Slot B: parked by ~24 s. Slot A: by ~28 s.
PARK +5

STRETCH (○)
OFF by default
Back to the spot, fires the GARDEN POLLEN only if the camera sees our CELL still **up**.

Times are from S1 with no start delay. A start delay makes it all later. From S2 it first drives about 5 s to the shot spot.

PARKING ALWAYS WINS: at the Park-by time (24 s, or earlier if the drive is long) it stops what it is doing and parks.

During AUTO: hands off

- **Don't touch** the robot or the gamepads. The gamepads do nothing in AUTO.
- **Don't press STOP.** All motors stop at once and the robot won't park. Emergencies only.
- It waits or skips a step? **Let it**: those are its own rules (middle column). After the match, tell the coach: the log has every step.
- Watch the `Step` line to see what it is doing.

After AUTO: TeleOp

1. The 30 s AUTO timer ends AUTO. The robot holds its park until then.
2. **BioBuzz TeleOp** is already selected for you: press **INIT**.
3. `Alliance` line: AUTO already set our colour. Check it anyway.
4. Field-centric on? The screen says `Start heading: from AUTO`. Don't press OPTIONS.
5. Don't touch the robot. ▶ **START** when TeleOp starts.

If this happens in AUTO

If...	the robot...
No CELL in view	fires the pre-loads straight ahead.
Our CELL already down	never fires into it (G417). No GARDEN; it parks.
Balls left after shooting (jam)	no GARDEN (G407: max 4 balls); it parks.
Blocked 1 s (e.g. by our partner)	drops that step and parks; while parking it tries again until AUTO ends.
The Pinpoint fails (low battery)	stops; backs off the wall (LEAVE). No PARK.

Practice: set our HIVE to the match start first (**our CELL up**). If our CELL is down, **AUTO won't shoot** (G417).

The Step line during AUTO

Step says	It is...
`SHOT SPOT · AIM … · FIRE …`	shooting the pre-loads
`GARDEN …`	in the GARDEN
`PARK: …`	driving to the slot
`PARKED B / PARKED A`	**done ✓**
`FALLBACK DONE (…)`	Pedro was not ready
`POSE LOST: stopped`	the Pinpoint failed

PRE-MATCH CHECKLIST

- ☐ **Partner agreed:** who has S1 · A or B · our GARDEN · our lane
- ☐ **Op mode:** BioBuzz AUTO RED / BLUE = our colour
- ☐ **Battery 12.5 V+** · **4 pre-loads**, one per pocket
- ☐ **Robot at S1 or S2:** intake against the wall, centred, square
- ☐ **INIT**, hands off · title = **our colour**
- ☐ **Start** and **Park slot** as agreed
- ☐ **Shoot ON, GARDEN ON** · stretch, delay, park by as agreed
- ☐ **No !! line** · **Route check: OK** · **Pedro: ready**
- ☐ **Place** line = where the robot is
- ☐ Last look, then ▶ **START** when the match starts. Hands off.